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Forecasting WTI Crude Oil Futures Returns and Volatility: An ARMA–GARCH Approach with Multivariate Analysis

Project Work

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Abstract

This paper studies the forecasting of West Texas Intermediate (WTI) crude oil futures returns and conditional volatility through a family of sparse ARMA-GARCH models, complemented by a multivariate lead-lag analysis with ten energy and financial instruments. Daily closing prices from July 2022 to April 2026 are merged into a panel of 1002 trading days and transformed into percentage log returns. Preliminary tests confirm the typical stylised facts of oil returns: stationarity of the series, strong ARCH effects, zero unconditional mean, excess kurtosis and negative skewness. An ARMA(6,6) specification with only the lag-2 and lag-6 coefficients retained, consistent with both the auto-ARIMA identification and the ACF/PACF inspection, is used for the conditional-mean modeling.

Eight competing specifications are compared by crossing four variance models (sGARCH, EGARCH, GJR-GARCH and GARCH-M) with two mean equations: a univariate ARMA and an ARMAx augmented with two exogenous regressors: the once-lagged Energy Select Sector ETF return (XLE) and the twice-lagged US Dollar Index return (USD), both identified via cross-correlation and Granger causality. All models are estimated on an 80/20 train-test split and evaluated through a recursive rolling forecast using metrics like RMSE and MAE for both mean and variance, and the Diebold-Mariano (DM) test of equal predictive accuracy. The selected specification, M6 = ARMAx(6,6)-eGARCH(1,1) with skewed Student-t innovations, achieves the cleanest standardised-residual diagnostics among the eight, with no residual ARCH effect at any tested lag. The multivariate analysis confirms XLE and USD as the most robust leading indicators of WTI returns among the ones considered in this work.

1 Introduction

Crude oil is arguably the most important commodity in the global economy. Its price feeds directly into the cost of transportation, manufacturing and heating, and indirectly influences inflation, monetary policy and financial markets. Among crude oil benchmarks, West Texas Intermediate (WTI) is the primary reference for North American grades and one of the two most liquid oil contracts in the world, together with Brent.

Accurately forecasting WTI returns is therefore of critical importance to a broad spectrum of economic agents, ranging from energy producers and refining companies managing operational exposure, to financial institutions quantifying market risk, and policymakers monitoring inflationary pressures.

The inherent difficulty of forecasting crude oil prices is well established in the literature. Energy markets are subject to a unique confluence of structural forces: supply constraints, geopolitical disruptions, macroeconomic cycles and financial contagion that generate return distributions characterized by severe non-normality, volatility clustering, and asymmetric responses to market shocks. These stylized facts render classical homoskedastic linear models fundamentally inadequate for capturing the full distributional dynamics of WTI returns, motivating the extensive body of research on conditional heteroskedasticity models and their extensions to energy commodities

(Kang and Yoon, 2013; Wei et al., 2010).

The sample period considered in this study, spanning from July 2022 to April 2026, is of particular empirical relevance. It encompasses a sequence of geopolitical shocks of exceptional severity that collectively reshaped global energy market dynamics. The Russian invasion of Ukraine in February 2022 generated the largest supply-side disruption in the global oil market since the 1990 Gulf War, triggering a structural reconfiguration of international crude trade flows and sustaining elevated price volatility throughout 2022. Subsequent escalations in the Middle East, including the outbreak of the Israel-Hamas conflict in October 2023, Houthi attacks on Red Sea maritime traffic, and the direct Iran-Israel confrontations of 2024, introduced persistent risk premia into forward energy prices. Most critically, the partial closure of the Strait of Hormuz in 2026, a chokepoint through which approximately 20% of globally traded crude oil transits, produced an immediate non-linear price response accompanied by an explosion in realized volatility. This sequence of events provides a challenging empirical environment in which to evaluate the robustness and adaptability of alternative forecasting specifications.

Against this backdrop, the present study develops and evaluates a progressive econometric framework for one-step-ahead forecasting of WTI futures returns. The methodology proceeds through three sequential stages of increasing complexity. In the first stage, a restricted ARMA specification is identified and estimated for the conditional mean of WTI log returns, following standard pre-estimation diagnostics including unit root testing, serial correlation analysis, and assessment of the unconditional mean. In the second stage, diagnostic evidence of conditional heteroskedasticity in the ARMA residuals motivates the transition to a Generalized Autoregressive Conditional Heteroskedasticity (GARCH) framework for the conditional variance. A comprehensive suite of competing specifications is evaluated, with the Exponential GARCH (EGARCH) model selected on the basis of its ability to capture asymmetric leverage effects without imposing non-negativity constraints, paired with a Skewed Student-t distributional assumption to accommodate the leptokurtosis and directional asymmetry empirically documented in WTI returns. In the third stage, the univariate framework is extended to a multivariate ARMAX-GARCH specification incorporating exogenous leading indicators identified through Cross-Correlation Function analysis. The Energy Select Sector ETF (XLE) and the U.S. Dollar Index (USD) are retained as statistically significant predictors at lags -1 and -2, respectively, consistent with established price discovery mechanisms and the inverse oil-dollar relationship documented in the empirical finance literature (Reboredo, 2012).

Out-of-sample forecasting performance is evaluated across eight competing GARCH-class specifications using a rolling recursive window over the last 20% of observations. Predictive accuracy is assessed through Mean Squared Error, Mean Absolute Error, and the Patton QLIKE loss function, the latter chosen for its robustness to noise in variance proxies and its asymmetric penalization of volatility under-prediction, a property of particular relevance in risk management applications. Empirical results indicate that asymmetric multivariate ARMAX-EGARCH specifications with Skewed Student-t innovations outperforms both symmetric and univariate benchmarks.

The remainder of this paper is organized as follows. Section 2 reviews the relevant literature

on crude oil price forecasting. Section 3 describes the data sources, construction, and preliminary descriptive analysis. Section 4 presents the econometric methodology in detail. Section 6 reports and discusses the empirical results. Section 7 concludes with a summary of the main findings, a discussion of the limitations of the study, and directions for future research.

2 Literature Review

Forecasting crude oil prices, particularly West Texas Intermediate (WTI) futures, has been extensively explored due to the commodity's volatile nature and its critical role in the global economy and geopolitical shocks. Historically, researchers have relied on traditional linear time series frameworks to capture the conditional mean behavior of commodity prices. For example, Box-Jenkins methodologies, such as ARIMA and ARMA models, have been widely utilized to establish baseline forecasts for various raw materials, ranging from agricultural goods to crude oil series. However, while these univariate models provide a strong statistical foundation for estimating expected values, a major shortcoming is their assumption of homoskedasticity. To enhance the predictive accuracy of these baseline models without formally addressing variance, a number of authors have turned to computational algorithms. In fact, some researchers have tried to combine econometric models to neural networks and machine learning techniques to enhance the predictive accuracy of these baseline models. Studies exploring hybrid ARIMA-ANN models have demonstrated that combining linear econometric methods with artificial neural networks can successfully capture complex non-linear patterns. Similarly, dos Santos et al. (2024) demonstrate that utilizing computational algorithms, such as Particle Swarm Optimization, to combine traditional linear models into ensembles significantly improves one-step-ahead forecasting for absolute WTI prices. Yet, while these algorithmic enhancements reduce error mathematically, they act as opaque models that fail to structurally explain the underlying economic realities driving daily, high-frequency price fluctuations.

de Souza e Silva et al. (2010a) tried to predict prices of oil by considering high-frequency daily fluctuations as unpredictable noise by using Wavelet Analysis and then the Hidden Markov Model (HMM). However using this strategy in a financial environment is not the best choice since the noise left out is the actual description of the shocks affecting oil prices. As a result, we decided to analyze the effect of market shocks on WTI returns by applying GARCH models to our time series.

A primary structural reality of energy markets is severe volatility clustering and susceptibility to sudden market shocks. Because linear and ensemble models struggle with this complexity, some literature attempts to improve forecasting accuracy by artificially smoothing the underlying data. For instance, de Souza e Silva et al. (2010b) successfully utilized wavelet analysis paired with Hidden Markov Models to predict multi-day oil price trends by deliberately filtering out high-frequency daily price movements as "noise". However, using this strategy in a financial environment is not the best choice since the noise left out is the actual description of the shocks affecting oil prices. As a result, we decided to analyze the effect of market shocks on WTI returns by applying conditional heteroskedasticity models (such as GARCH) to our time series. To formally address volatility persistence rather than discarding it, Kang and Yoon (2013) found

that advanced, long-memory conditional heteroskedasticity models, such as FIGARCH, are highly effective at capturing petroleum market turbulence. On the other hand, an important weakness in these standard GARCH-class frameworks is that they frequently treat price shocks symmetrically. Because energy markets exhibit strong leverage effects, where unexpected negative news impacts volatility much more aggressively than positive news, this study departs from symmetric and linear assumptions. Instead, we transition to an EGARCH (Exponential GARCH) framework to explicitly capture these crucial asymmetric leverage effects in the daily returns of WTI futures.

Furthermore, while univariate ARMA and EGARCH models effectively isolate an asset's historical price and variance behaviors, they treat the asset independently, ignoring broader macroeconomic realities. Recognizing the necessity of external information, recent literature has heavily favored multidimensional approaches. Coppola (2008) utilized Vector Error Correction Models (VECM) to exploit the cointegrating relationship between spot and futures prices, successfully outperforming random walk benchmarks by incorporating cross-market data. Similarly, alternative approaches have proven the value of mixed-frequency data and textual sentiment indicators to capture external market drivers. Building upon this consensus that energy prices are dynamically interlinked with external factors, this project ultimately transitions to a multivariate econometric framework. By implementing a Vector Autoregression (VAR) model for the conditional mean paired with a Dynamic Conditional Correlation (GARCH-DCC) for the conditional variance, this study addresses the shortcomings of isolated univariate forecasting. This progression allows for a highly granular, one-step-ahead forecast of end-of-week WTI returns that properly accounts for both asymmetric volatility and observable cross-market macroeconomic dependencies.

3 Data

3.1 Data description and sources

The empirical analysis uses a comprehensive dataset with a daily frequency chosen to capture both the dynamics of the crude oil market and its possible macroeconomic dependencies.

The focus of this paper is the West Texas Intermediate (WTI) crude oil futures contract which is the premier benchmark for global energy pricing. The WTI series includes the official daily settlement price calculated by the NYMEX (CME Group), which is determined by trading activity during the final window prior to the session's close. WTI was deliberately selected over other benchmarks due to the deep liquidity of its futures market, which minimizes microstructural noise and makes it the optimal candidate for volatility modeling and GARCH specifications.

Subsequently, to construct the multivariate framework and capture cross-market dependencies, the WTI series is augmented with a selected group of exogenous macroeconomic variables. These supplementary variables include sector-specific equities (XLE ETF), broad market indices (S&P 500), currency benchmarks (the U.S Dollar Index), volatility indices (VIX) and related physical commodity metrics (Baltic Dry Index, Carbon EUA, Heating Oil and RBOB Gasoline).

Despite their heterogeneous asset classes, each dataset strictly adheres to a common tabular

format comprising standard trading metrics (Date, Open, High, Low, Price, Volume, and Percentage Change). Crucially, the 'Price' variable denotes the official closing valuation for the respective trading session. For futures contracts, this represents the official settlement price calculated by the exchanges (CME, ICE, COMEX, NYMEX) based on the final trading window prior to the session close. Conversely, for equities (XLE) it reflects the closing price on NYSE Arca, for the VIX it relies on the CBOE closing fixing, and for the BDI it denotes the afternoon fixing of the Baltic Exchange in London.

The resulting sample period spans from July 2022 to April 2026. This specific timeframe was purposefully selected, as it coincides with the window capturing the decade's most significant geopolitical crude oil price shocks, as already mentioned in the introduction (Section 1).

The final dataset is built by merging daily closing prices for the eleven time series summarised in Table 1.

Table 1: Variables used in the empirical analysis.

Ticker	Description	Asset class
WTI	Crude Oil WTI Futures	Energy (endogenous)
RBOB	Gasoline RBOB Futures	Energy
NYFM6	Heating Oil Futures	Energy
XLE	Energy Select Sector SPDR ETF	Energy equities
CFI2M6	EUA Carbon Emissions Futures	Environmental
BDI	Baltic Dry Index	Shipping / Cycle
VIX	CBOE Volatility Index (VIX)	Risk sentiment
USD	US Dollar Index Futures	FX / Macro
GC	Gold Futures	Safe haven
HG	Copper Futures	Industrial metal
ESM26	S&P 500 Futures	Equity benchmark

WTI is the endogenous target variable of the study; the remaining ten series act as candidate explanatory or leading variables in the multivariate analysis of Section 4.3 and as a pool from which the exogenous regressors of the ARMAx specifications (M5–M8) are drawn. Each of them captures a distinct transmission channel that can plausibly affect crude oil returns: from direct refining co-movements (RBOB, NYFM6) to financial conditions (VIX, USD, ESM26), macroeconomic activity (BDI, HG) and safe-haven demand (GC). A more detailed description of the datasets is reported below:

- **WTI** (West Texas Intermediate Crude Oil Futures). These are the NYMEX WTI light sweet crude oil futures, a 1,000-barrel contract with delivery at Cushing, Oklahoma. WTI is the primary pricing benchmark for North American crude oil and one of the two most liquid oil contracts in the world, together with Brent. It is classified as a light sweet crude, making it particularly suitable for refining into gasoline and distillates.
- **RBOB** (Reformulated Blendstock for Oxygenate Blending). These are RBOB gasoline futures, a 42,000-gallon NYMEX contract with delivery at New York Harbor. RBOB is the raw material that refineries sell to blenders, who add ethanol to produce commercial

gasoline at the pump.

- **NYFM6** (Heating Oil Futures). It is a key "refined benchmark" of the New York Mercantile Exchange (NYMEX). NYFM6 measures the wholesale price of heating oil, derived directly from the physical refining of crude oil, and it is expressed in dollars per gallon.

- **XLE** (Energy Select Sector SPDR Fund). It is an equity ETF launched by State Street in 1998, ticker XLE on NYSE Arca, which tracks the Energy Select Sector Index of the S&P 500. It is the primary passive vehicle for expressing a view on the US oil & gas industry.

- **CFI2M6** (EU Carbon Emission Allowance Futures). These are EUA (European Union Allowance) emission permit futures from the EU ETS (Emissions Trading System), listed on ICE Endex (originally ECX) in Amsterdam. One EUA grants the holder the right to emit one metric ton of CO_2 equivalent. The rationale for including them was twofold: the price of CO_2 is part of the marginal cost for European refineries and thus impacts refining margins and a rising carbon price accelerates the substitution of crude oil with renewables and is expected to weigh on structural oil demand.

- **BDI** (Baltic Dry Index). It is a composite index published by the Baltic Exchange in London that measures the cost of maritime shipping for dry bulk cargoes (iron ore, coal, grain, bauxite). Historically, it has been considered a leading indicator of global industrial activity: if world trade accelerates, bulk carriers become saturated, and freight rates rise before the real effects show up in official statistics. Hence its presumed usefulness for forecasting crude oil demand.

- **VIX** (It is the famous "fear gauge" of the Chicago Board Options Exchange). The VIX measures the 30-day implied volatility of the S&P 500, extracted from the prices of packages of put and call options at various strikes, according to a model-free methodology published by the CBOE in 2003. It is expressed in annualized percentage points (a VIX of 20 implies an expected volatility of the S&P 500 of 20% annualized over the next 30 days).

- **USD** (U.S. Dollar Index Futures). These are futures on the DXY, the US dollar index against a basket of six currencies with the following weights: Euro 57.6%, Japanese Yen 13.6%, British Pound 11.9%, Canadian Dollar 9.1%, Swedish Krona 4.2%, and Swiss Franc 3.6%.

- **GC** (Gold Futures). These are the COMEX (a division of the NYMEX-CME) GC futures, a 100-troy-ounce contract with delivery in 100 oz or 1 kg bars. Gold is classically included in any commodity system because it is considered an inflation hedge (expected inflation also impacts crude oil), a safe haven in opposition to the dollar, and a proxy for global monetary liquidity.

- **HG** (Copper Futures). These are the COMEX HG futures, a 25,000-pound contract for high-purity copper. Its demand is heavily dominated by construction, electrical wiring, automotive (increasingly so with the EV transition), and industrial electronics, making it one of the best real indicators of global industrial activity, particularly in the Chinese

economy.

- **ESM26** (S&P 500 Futures). These are the CME E-mini S&P 500 futures, the most liquid equity contract in the world, with the primary American large-cap index as its underlying asset.

3.2 Preliminary data analysis

Figure 1 plots the WTI log price level and the corresponding percentage log returns over the full sample. The upper panel reveals a persistent downward drift from mid-2022 onwards, interrupted by sharp reversals corresponding to the major geopolitical events described in Section 1. The lower panel exhibits the characteristic features of high-frequency financial returns: volatility clustering, occasional large spikes, and no discernible trend in the mean.

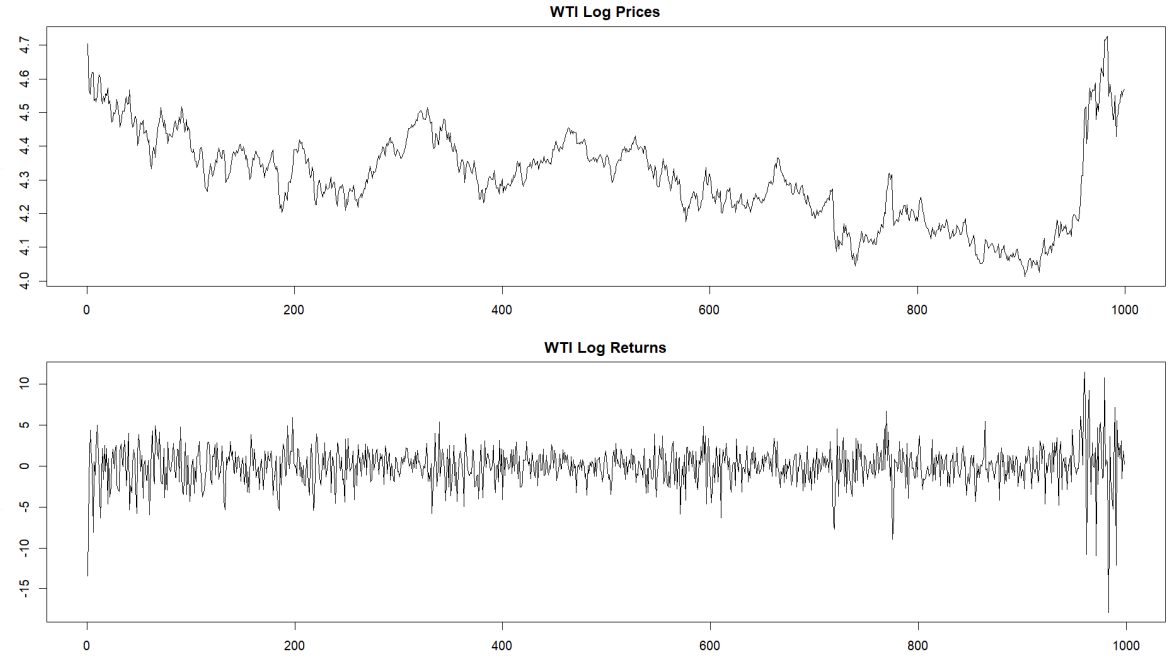


Figure 1: WTI log prices and percentage log returns, July 2022 – April 2026.

Table 2 reports the descriptive statistics of WTI percentage log returns. The stylized facts typical of financial returns, namely non-normality and heavy tails, are clearly evidenced in our sample, as the return series exhibits leptokurtosis and negative skewness.

Table 2: Descriptive statistics — WTI percentage log returns (daily, $T = 1002$).

Mean	Median	Std. dev.	Min	Max	Skewness	Ex. kurt.
−0.0136	0.0840	2.3636	−17.9298	11.5187	−0.8127	9.7078

4 Methodology

This section provides a description of the methodology strategy to predict closure price of WTI Futures. Given the complex dynamics related to the data and the instability of geopolitical situations, forecasting WTI prices has required progressive modeling approaches. Methodology frameworks can be divided into three sequential steps: Univariate Mean Modeling (ARMA); Volatility Modeling (GARCH); Multivariate models (VAR, ARMAX).

4.1 ARIMA modeling

The first phase of the methodology employs a univariate linear autoregressive model. However, before estimating the model, fundamental diagnostic tests have been applied to identify structural components and ensure that the chosen model is appropriate for the analyzed time series. First of all, the Augmented Dickey-Fuller (ADF) (Table 3) test has been performed to check for the presence of a unit root in order to determine whether the time series is stationary or not. As expected, log prices are not stationary, however, the analysis on the log returns confirmed that no unit roots were detected.

Table 3: Augmented Dickey-Fuller (ADF) Test Results for WTI Log Returns (r_{wti})

Series	**Type**	**Lag Order**	**Dickey-Fuller**	**p-value**
Log Prices ($\ln P_t$)	Constant + Trend	15	-1.5456	0.7707
Log Returns (r_t)	Constant + Trend	15	-8.6102	0.0100

Subsequently, a T-test was performed figuring out that the unconditional mean of the WTI (Table 4) returns was statistically insignificant and so equal to zero.

Table 4: One-Sample t-test on WTI Log Returns (r_{wti})

Variable	**Mean**	***t*-stat**	**df**	***p*-value**	**95% Conf. Interval**
Log Returns (r_t)	-0.0136	-0.1820	997	0.8556	[-0.1604, 0.1332]

Inspection of the ACF and PACF correlograms, (Figure 2), revealed significant serial correlation at lags 2, 6, 8 and 12, informing the identification of the optimal ARMA order.

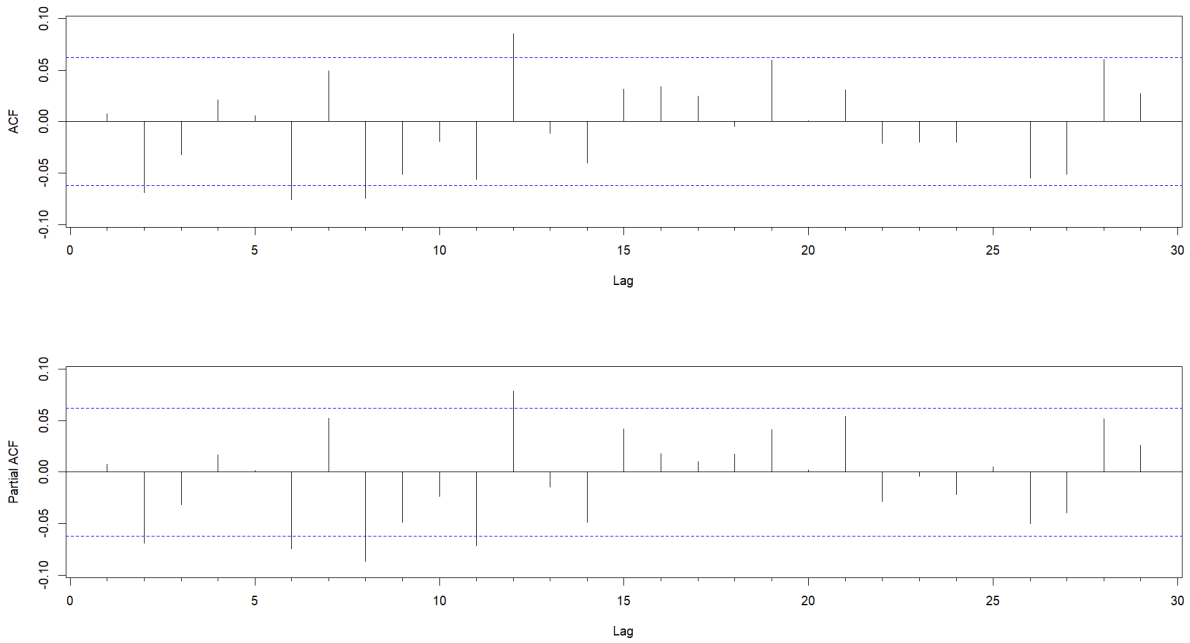


Figure 2: WTI Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF)

To capture the linear serial dependence within the stationary log returns, an Autoregressive Moving Average (ARMA) framework was employed. The general ARMA (p,q) for the WTI log returns at time t (r_t) is specified as:

$$r_t = \phi_2 r_{t-2} + \phi_6 r_{t-6} + \varepsilon_t + \theta_2 \varepsilon_{t-2} + \theta_6 \varepsilon_{t-6} \tag{1}$$

The selection of the optimal lag lengths (p and q) is a critical step to balance model accuracy and simplicity. Initially, an automated selection process relies on Information Criteria (such as the Akaike Information Criterion, AIC) to suggest a baseline global order. However, fitting a high-order ARMA model often results in over-parameterization, where several intermediate lags carry no statistical significance. To refine the model and avoid overfitting, we estimate a custom, restricted ARMA specification. Using Maximum Likelihood Estimation, we actively constrain the coefficients of statistically insignificant intermediate lags to exactly zero, isolating only the specific historical lags that meaningfully drive current WTI prices.

Table 5: Custom Sparse ARIMA (6,0,6) Model Results for WTI Log Returns (r_{wti})

Parameter	**Estimate**	**Std. Error**	***z*-value**	***p*-value**	**Significance**
ar2	0.5610	0.2031	2.7623	0.0057	**
ar6	-0.4063	0.1388	-2.9272	0.0034	**
ma2	-0.6074	0.2034	-2.9865	0.0028	**
ma6	0.3233	0.1480	2.1835	0.0290	*
Model Fit Statistics					
AIC: 4542.11	BIC: 4566.63	Log Likelihood: -2266.05		RMSE: 2.3434	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' '

Finally, to validate the adequacy of the selected model (Table 5), post-estimation diagnostic tests have been performed on residuals.
The Ljung-Box test is applied to verify that no significant autocorrelation remains in the residuals. Subsequently, the Jarque-Bera test is used to assess the normality of the error distribution and the ARCH-LM test is conducted to detect the presence of conditional heteroskedasticity. In a financial time series, while the ARMA model successfully filters out linear dependence, the residuals typically fail both the Jarque-Bera and ARCH tests, revealing heavy tails and volatility clustering (Table 6).

Table 6: Diagnostic Tests on Residuals of Custom ARIMA(6,0,6) Model

Test	**Lags**	**Statistic**	***p*-value**	**Conclusion**
Ljung-Box	10	10.885	0.3665	Fail to reject $H_0 \implies$ No autocorrelation
Jarque-Bera	–	1663.806	<0.001	Reject $H_0 \implies$ Non-normal distribution
ARCH-LM	15	204.680	<0.001	Reject $H_0 \implies$ Heteroskedasticity present

These specific diagnostic failures dictate the necessity of transitioning to following section, where we model the conditional variance.

4.2 GARCH modeling

Following the specification of the ARMA model to capture the conditional mean of WTI returns, diagnostic tests of the residuals (specifically the ARCH-LM test) revealed profound heteroskedasticity. Major global events, such as the outbreak of the Russia-Ukraine war and subsequent escalations in the Middle East, do not result in isolated, single-day price adjustments. Instead, they create structural 'regimes' of heightened market anxiety, where large shocks are followed by sustained periods of instability. To accurately model this dynamic risk premium, a transition to a Generalized Autoregressive Conditional Heteroskedasticity (GARCH) framework was required. To determine the optimal specification for capturing WTI volatility, a comprehensive suite of conditional heteroskedasticity models was evaluated. Several prominent frameworks (including ARCH, sGARCH, iGARCH, GARCH-M and GJR-GARCH) were systematically rejected due to their underlying mathematical assumptions fundamentally misaligned with the economic realities of the energy market. The basic ARCH and standard sGARCH models were the first ones to be excluded. The critical limitation of these symmetric baselines is their reliance on squared residuals, which assumes that positive and negative price shocks generate the exact same magnitude of future volatility. In the WTI market, this assumption is structurally invalid. Energy markets exhibit a pronounced "supply-side panic", wherein geopolitical threats restricting oil supply (price spikes) trigger significantly more turbulent trading environments than gradual price declines caused by oversupply. Because ARCH and sGARCH are agnostic to the direction of a shock, they fail to capture the market's disproportionate sensitivity to supply disruptions. The standard GARCH model specification is defined as:

$$\sigma_t^2 = \omega + \sum_{i=1}^{p} \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^{q} \beta_j \sigma_{t-j}^2 \tag{2}$$

Similarly, models that mischaracterized the duration and nature of market memory were discarded. The IGARCH model forces volatility persistence to equal exactly one, implying an infinite "half-life" for any given shock. However, the empirical findings indicate that while WTI volatility is highly persistent, it does not exhibit infinite memory. The I-GARCH model formula is defined as:

$$\sigma_t^2 = \omega + \sum_{i=1}^{p} \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^{q} \beta_j \sigma_{t-j}^2 \quad \text{with} \quad \sum_{i=1}^{p} \alpha_i + \sum_{j=1}^{q} \beta_j = 1 \tag{3}$$

The GARCH-M model was similarly rejected. Although standard in equity research, where investors demand a risk premium for holding volatile stocks, this specification is inappropriate for physical commodities. WTI pricing is driven by inventory constraints, convenience yields, and hedging operations, rather than the classic risk-return trade-offs characteristic of stock indices. The GARCH-M model formula is defined as:

$$\begin{aligned} r_t &= \mu + \lambda \sigma_t^2 + \varepsilon_t \quad \text{with} \quad \varepsilon_t | I_{t-1} \sim iid(0, \sigma_t^2) \\ \sigma_t^2 &= \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2 \end{aligned} \tag{4}$$

Subsequently, tGARCH (GJR-GARCH) were tested considering the need to account for the asymmetric "leverage effect". Although these models successfully introduce parameters to treat positive and negative shocks differently, they still mathematically model the variance in absolute terms. As a result, they require strict non-negativity constraints on their coefficients to prevent the variance from becoming negative. However, during the extreme market malfunctions observed in the sample, including unprecedented intraday reversals, these rigid constraints often force poor model optimization and severely restrict analytical flexibility. The tGARCH model is specified as follows:

$$\sigma_t^2 = \omega + \sum_{i=1}^{s} (\alpha_i + \gamma_i S_{t-i}^-) \varepsilon_{t-i}^2 + \sum_{j=1}^{m} \beta_j \sigma_{t-j}^2 \tag{5}$$

In light of these structural limitations, the Exponential GARCH (EGARCH) model was selected as the optimal framework. This specification effectively resolves the deficiencies inherent in both the symmetric baselines and alternative asymmetric models. The EGARCH model has the following specification:

$$\log(\sigma_t^2) = \omega + \sum_{j=1}^{q} \beta_j \log(\sigma_{t-j}^2) + \sum_{i=1}^{p} \alpha_i \left| \frac{\varepsilon_{t-i}}{\sigma_{t-i}} \right| + \sum_{k=1}^{p} \gamma_k \frac{\varepsilon_{k-i}}{\sigma_{k-i}} \tag{6}$$

Because the EGARCH model specifies the natural logarithm of the conditional variance, its parameters are unconstrained and carry no risk of generating mathematically impossible negative

variance. This logarithmic formulation enables the model to adapt robustly to extreme price movements and effectively capture the asymmetric dynamics of the energy sector. Consequently, it precisely quantifies the differential market impact of severe geopolitical supply threats compared to standard fluctuations.

To finalize the model specification, the statistical distribution of the residuals had to be addressed. While a standard assumption in classical econometrics is that market returns follow a Normal (Gaussian) distribution, preliminary Jarque-Bera tests conclusively rejected the null hypothesis of normality for WTI. Relying on a standard normal distribution for the 2022–2026 period would imply that the violent price swings, caused by the invasion of Ukraine or Middle Eastern escalations, are statistical impossibilities that should mathematically occur only once in a millennium. In reality, the modern energy market is heavily prone to these unpredictable events, necessitating a distribution that accounts for extreme tail risks.

To address this limitation, the EGARCH framework was paired with a Skewed Student-t distribution (sstd). This specification serves two critical empirical functions. First, the Student-t framework accounts for fat tails (excess kurtosis), allowing the model to accommodate a much higher frequency of extreme market shocks. Second, the skewness parameter captures the directional asymmetry of these extremes, acknowledging that a sudden geopolitical supply deficit generates a fundamentally different risk profile than a standard demand recession.

The final output of the selected EGARCH-sstd model reveals a highly significant persistence parameter, confirming a phenomenon that can be termed the "echo" effect of geopolitical risk. Even after the immediate shock of a conflict subsides, the mathematical memory of that event dictates that the risk premium remains embedded within WTI prices for weeks.

Table 7: ARMA–eGARCH full parameter table

Parameter	**Estimate**	**Std. Error**	***t*-stat**	***p*-value**	**Sig.**
Mean equation					
$\hat{\phi}_2$	0.8865	0.1366	6.4915	0.0000	***
$\hat{\phi}_6$	−0.4909	0.0589	−8.3353	0.0000	***
$\hat{\theta}_2$	−0.9269	0.1453	−6.3803	0.0000	***
$\hat{\theta}_6$	0.4746	0.0583	8.1347	0.0000	***
Variance equation					
$\hat{\omega}$	0.0243	0.0031	7.8707	0.0000	***
$\hat{\alpha}_1$	−0.0342	0.0202	−1.6978	0.0895	n.s.
$\hat{\beta}_1$	0.9811	0.0001	9574.0198	0.0000	***
$\hat{\gamma}_1$	0.0829	0.0053	15.6052	0.0000	***
Distribution parameters					
skew	0.8005	0.0423	18.9411	0.0000	***
shape (ν)	8.1375	1.8990	4.2851	0.0000	***

Significance: ***$p < 0.001$, **$p < 0.01$, *$p < 0.05$, n.s. – not significant.

GARCH model: eGARCH(1,1). Distribution: sstd.

This model has the following specification:

$$
\begin{aligned}
r_t &= \phi_2 r_{t-2} + \phi_6 r_{t-6} + \theta_2 \varepsilon_{t-2} + \theta_6 \varepsilon_{t-6} + \varepsilon_t, \\
\log(\sigma_t^2) &= \omega + \sum_{j=1}^{q} \beta_j \log(\sigma_{t-j}^2) + \sum_{i=1}^{p} \alpha_i \left| \frac{\varepsilon_{t-i}}{\sigma_{t-i}} \right| + \sum_{k=1}^{p} \gamma_k \frac{\varepsilon_{k-i}}{\sigma_{k-i}}
\end{aligned} \tag{7}
$$

4.3 Multivariate modeling

While univariate models effectively capture the internal dynamics of a single time series, they inherently fail to account for the broader financial ecosystem. Consequently, this section explores the interdependence between WTI crude oil and a curated set of financial indicators. The primary objective is to transition from a purely descriptive analysis to a predictive framework by identifying variables that exhibit a statistically significant predictive lead over WTI returns.

The transition from a univariate to a multivariate framework necessitates a rigorous approach to data integrity. Prior to conducting any statistical inference, the inherent asynchronicity of global financial markets had to be addressed. This temporal mismatch constitutes the primary challenge when combining data from the NYMEX (WTI), NYSE (XLE), and Forex markets (USD), given that each exchange operates according to distinct holiday calendars and trading hours.

Consequently, the first methodological step involved the construction of a balanced dataset through a systematic synchronization protocol. A master timeline was generated, encompassing every trading date present across the original time series. This approach ensured that unique, market-specific information was preserved rather than discarded during the data aggregation process.

To solve the data gaps created by regional market closures, the Last Observation Carried Forward (LOCF) method was employed. By propagating the last available settlement price to the subsequent day, it is assumed that—in the absence of new trading volume—the market's "last known state" remains the most accurate proxy for value. Furthermore, a backward fill (Next Observation Carried Backward) was applied exclusively to the first observation of the sample to stabilize the starting point and enable the calculation of the initial return.

A critical methodological decision involved the explicit rejection of alternative imputation techniques, such as mean interpolation or linear splines. This choice is underpinned by strict econometric rationales that are fundamental to predictive modeling:

- Elimination of Look-Ahead Bias: mean interpolation is inherently bi-directional, utilizing information from time $t+1$ to estimate a missing value at time t. In a forecasting framework, this violates the martingale property of the information set by inadvertently "leaking" future data into the historical sample.
- Market Realism and Volatility Integrity: financial prices evolve as stochastic processes.

Interpolating between distant price points synthesizes artificial data, which spuriously dampens the estimated variance. The LOCF method ensures the model does not fabricate price movements or volatility where no actual trading occurred.

- Lead-Lag Preservation: given the objective to identify whether specific indicators lead WTI returns, LOCF serves as the most conservative baseline. It ensures that any detected correlation at a specific lag reflects genuine market transmission rather than a mathematical artifact introduced by smoothing algorithms.

Following the synchronization of the dataset and the imputation of missing values, a statistical screening of the eleven candidate series was conducted.

The selection criterion was strictly empirical. To build the final ARMAX-GARCH model, we first conducted preliminary analyses to identify candidate variables. Specifically, we developed a simple ARMA model to understand the autoregressive-moving average dynamics of the dependent variable, and estimated a VAR model to capture the interdependencies among potential explanatory variables. From these preliminary analyses, we extracted the statistically significant coefficients from both the ARMA specification and the VAR equation describing WTI.

As a result of the preliminary Cross-Correlation Function analysis, only three variables, the ones detailed below, demonstrated statistically significant leading properties. The remaining candidate series were systematically excluded from the predictive multivariate framework, as they did not exhibit statistical significance at a daily frequency.

Based on the VAR restricted model results presented in Table 8, we initially selected three exogenous variables due to their statistical significance: XLE (Energy Select Sector SPDR ETF), HG (copper prices), and USD (US Dollar Index). However, following a certain period, the dataset was updated on a weekly basis. Consequently, we refined our variable selection and reduced the final set to two variables, XLE and USD, which continued to exhibit robust statistical significance and maintained their predictive power across the updated sample period.

Table 8: VAR restricted model – WTI equation

Parameter	**Estimate**	**Std. Error**	***t*-stat**	***p*-value**	**Sig.**
$r_{\text{WTI},t-1}$	-0.0614	0.0391	-1.570	0.1167	n.s.
$r_{\text{XLE},t-1}$	0.1363	0.0624	2.185	0.0291	*
$r_{\text{USD},t-1}$	0.2416	0.1696	1.425	0.1546	n.s.
$r_{\text{HG},t-1}$	0.1078	0.0451	2.390	0.0170	*
$r_{\text{WTI},t-2}$	-0.0745	0.0310	-2.404	0.0164	*
$r_{\text{USD},t-2}$	0.4112	0.1590	2.587	0.0098	**
Residual standard error: 2.305 on 990 degrees of freedom					
R^2: 0.0235 Adjusted R^2: 0.0176					
F-statistic: 3.978 on 6 and 990 DF, p-value: 0.0006097					
Significance: ***$p < 0.001$, **$p < 0.01$, *$p < 0.05$, n.s. – not significant.					

Subsequently, a variable was incorporated into the final ARMAX model only if it demonstrated a statistically significant leading relationship with the dependent variable, evidenced by negative lags in the Cross-Correlation Function (CCF). Finally, we extended this specification to include a

GARCH component to model conditional heteroskedasticity.

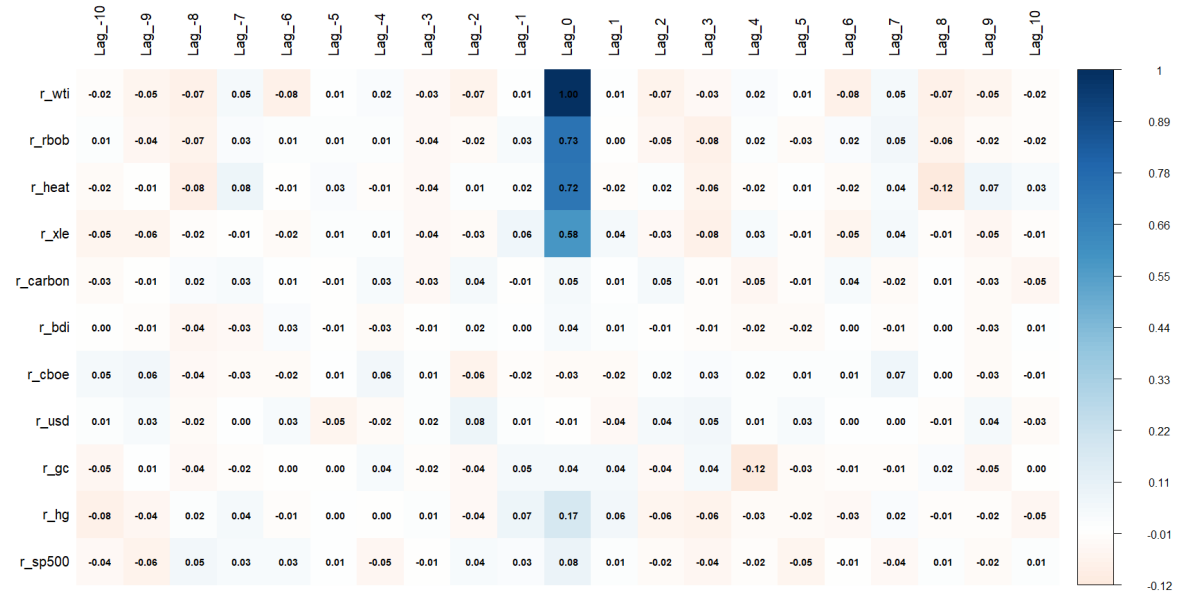


Figure 3: CCF Matrix from lag -10 to lag +10

Rigorous model specification requires the exclusion of variables that provide no incremental forecasting value. Consequently, assets such as RBOB Gasoline, Heating Oil, and the S&P 500 were omitted; despite exhibiting high correlations, they demonstrated strictly lag-zero relationships. Driven by arbitrage mechanisms like crack spreads, these assets evolve contemporaneously with WTI, reflecting the same information set without generating a predictive lead signal. Furthermore, variables such as Gold and the Baltic Dry Index were discarded due to evidence of reverse causality (e.g., WTI acting as the leading indicator) or an absence of detectable correlation at a daily frequency.

5 Forecasting

To simulate real-world trading conditions and prevent look-ahead bias, a rolling forecast methodology is implemented on the test set, which comprises the final 20% of the observations. Rather than maintaining static parameters throughout the testing period, a "recursive" window is employed, wherein the model is refitted multiple times. This periodic refitting allows the parameter estimates to adapt to structural breaks and evolving market regimes. The models generate one-step-ahead predictions for both the conditional mean and the conditional volatility across the entire test set.

The ultimate test of any econometric model does not lie in its in-sample fit but in its predictive accuracy on unseen data. To evaluate forecasting capabilities, the analysis compares a formalized suite of eight distinct model specifications. This suite is structurally divided into two primary categories: purely univariate models (M1–M4), which rely solely on the historical returns of WTI, and multivariate ARMAX models (M5–M8), which integrate the previously identified leading indicators (the Energy Sector ETF and the U.S. Dollar Index) into the conditional mean equation. Within these two broad categories, various variance dynamics are tested. These include standard

symmetric models (sGARCH), asymmetric models designed to capture leverage effects (eGARCH and gjrGARCH), and specifications that test the risk-return trade-off (GARCH-in-mean). To accommodate the severe non-normality identified during diagnostic testing, all eight models use the Skewed Student-t (sstd). Table 9 summarizes the eight model specifications defined at this point of the analysis.

Table 9: Competing model specifications.

Model	Variance equation	Mean equation	xreg
Panel A: Univariate			
M1	sGARCH	ARMA(2,6)	–
M2	EGARCH	ARMA(2,6)	–
M3	GJR-GARCH	ARMA(2,6)	–
M4	sGARCH-M	ARMA(2,6)	–
Panel B: With exogenous regressors			
M5	sGARCH	ARMAX(2,6)	XLE, HG, USD
M6	EGARCH	ARMAX(2,6)	XLE, HG, USD
M7	GJR-GARCH	ARMAX(2,6)	XLE, HG, USD
M8	sGARCH-M	ARMAX(2,6)	XLE, HG, USD

All models estimated with a skewed Student-t (sstd) error distribution.

The standard approach of comparing point forecasts is conducted using Mean Squared Error (MSE) and Mean Absolute Error (MAE). Because squared returns are notoriously noisy proxies, the analysis incorporates the Patton QLIKE loss function alongside MSE and MAE. QLIKE is particularly valuable in financial applications because it is robust to noise in the proxy and asymmetrically penalizes models that under-predict volatility, a crucial feature for risk management where unexpected variance can lead to severe capital losses. Table 10 reports the metrics evaluated on the test set. This evaluation framework guides the final selection of the best model, alongside the diagnostic analysis of the standardised residuals, which assesses the statistical adequacy of each specification through tests for remaining autocorrelation and conditional heteroskedasticity, which are dealt with in section 7.

Table 10: Out-of-sample rolling forecast losses, $n_{test} = 200$.

Model	$\mathbf{RMSE_{mean}}$	$\mathbf{MSE_{mean}}$	$\mathbf{MAE_{mean}}$	$\mathbf{QLIKE_{mean}}$	$\mathbf{MSE_{vol}}$	$\mathbf{MAE_{vol}}$	$\mathbf{QLIKE_{vol}}$
M1	3.1286	9.7882	2.0025	2.8632	805.29	10.1451	8001.84
M2	3.1161	9.7100	2.0014	3.0822	850.17	9.4488	372.69
M3	3.1287	9.7887	2.0030	2.9109	817.16	10.0618	2548.40
M4	3.1442	9.8860	2.0314	2.8699	781.65	10.2123	5667.10
M5	3.1135	9.6941	1.9989	2.8663	810.96	10.0104	4680.73
M6	3.1116	9.6819	1.9949	3.0694	899.60	9.4784	3808.07
M7	3.1043	9.6369	1.9939	2.8565	861.52	9.9734	150.98
M8	3.1224	9.7495	2.0111	2.8752	804.36	10.0616	3552.77

$\text{MSE}_{\text{mean}}/\text{MAE}_{\text{mean}}$: squared/absolute return forecast errors. $\text{MSE}_{\text{vol}}/\text{MAE}_{\text{vol}}$: squared/absolute volatility forecast errors (proxy: $\hat{\varepsilon}_t^2$). $\text{QLIKE}_{\text{mean}}$: Patton QLIKE loss.

Based on the results presented in Table 10, the analysis demonstrates that multivariate ARMAX-GARCH models (M5–M8) incorporating macroeconomic indicators marginally outperform univariate benchmarks on the test set, with Model M7 achieving the lowest mean squared error (MSE = 9.6369). While the performance gains are modest, they confirm that external financial spillovers provide incremental predictive value for WTI forecasting. The rolling forecast methodology with periodical refitting successfully captures evolving market regimes.

6 Results

6.1 Model Specification analysis

Table 11 aggregates the residual diagnostics for each of the eight fits: the weighted Ljung-Box test on standardised residuals (no remaining autocorrelation in the mean), the weighted Ljung-Box test on standardised *squared* residuals (no remaining autocorrelation in the variance), the weighted ARCH-LM test on standardised residuals at lag 5 and lag 7 (no remaining ARCH effects), the joint Sign Bias test, and the joint Nyblom statistic on parameter stability.

Table 11: Consolidated Model Comparison: Information Criteria and Residual Diagnostics

Model	**AIC**	**BIC**	**LogLik**	**LB std resid**	**LB^2 [9] p-val**	**ARCH [5] p-val**	**ARCH [7] p-val**	**Sign Bias joint p**	**Nyblom joint**
ARMA Group									
M1 sGARCH	4.2367	4.2895	-1681.46	✓	0.020 ×	0.004 ×	0.003 ×	0.315 ✓	1.259 ✓
M2 eGARCH	4.2422	4.3008	-1682.62	✓	0.010 ×	0.006 ×	0.004 ×	0.548 ✓	1.673 ✓
M3 gjrGARCH	4.2389	4.2976	-1681.32	✓	0.022 ×	0.004 ×	0.004 ×	0.378 ✓	1.794 ✓
M4 ARCH-M	4.2390	4.2977	-1681.35	✓	0.018 ×	0.003 ×	0.003 ×	0.297 ✓	1.295 ✓
ARMAx Group									
M5 sGARCH+x	4.2136	4.2842	-1665.02	✓	0.754 ✓	0.387 ✓	0.370 ✓	0.496 ✓	1.846 ✓
M6 eGARCH+x	4.2193	4.2957	-1666.28	✓	0.183 ✓	0.172 ✓	0.063 ✓	0.760 ✓	2.111 ✓
M7 gjrGARCH+x	4.2160	4.2924	*-1664.97*	✓	0.733 ✓	0.399 ✓	0.347 ✓	0.528 ✓	2.256 ✓
M8 ARCH-M+x	4.2160	4.2924	-1664.95	✓	0.726 ✓	0.353 ✓	0.340 ✓	0.460 ✓	1.881 ✓

This table contains the single most important finding of the in-sample analysis. All four univariate models (M1–M4) failed the variance diagnostics: the weighted Ljung-Box test on squared standardised residuals at lag 9 rejects the null at the 5% level (p-values 0.010-0.022), while the ARCH-LM tests at lags 5 and 7 provide even stronger evidence against homoskedasticity (p-values 0.003-0.006). This indicates that the GARCH equations of the univariate specifications fail to fully absorb the conditional heteroskedasticity present in the residuals, constituting a fundamental misspecification in the variance equation. By contrast, *all four ARMAX models (M5–M8) pass all variance diagnostics cleanly*, with p-values comfortably above conventional thresholds. Including the exogenous regressors in the mean equation has absorbed variance dynamics that the pure GARCH structure could not capture on its own. This is the strongest internal reason to prefer the ARMAX family over the purely univariate one.

All eight models pass the mean serial-correlation test (weighted LB on standardised residuals, $p \geq 0.93$ throughout) and the joint Sign Bias test, meaning that no model exhibits remaining autocorrelation in the mean nor asymmetric news-impact issues beyond what is already modelled. The Nyblom joint statistic does not exceed its 5% critical value for any of the models, indicating

that parameter estimates are stable over the sample period.[1]

6.2 Comparison of Multivariate Models & Model Selection

The diagnostic comparison narrows the choice to the ARMAx family, and within that family the in-sample IC differences are negligible. Table 12 reports the full optimal parameters of M6, the selected specification, with both standard and robust standard errors.

Table 12: M6 full parameter table

Parameter	Estimate	Std. Error	*t*-stat	*p*-value	Sig.
Mean equation					
$\hat{\phi}_2$	0.8687	0.0490	17.7181	0.0000	***
$\hat{\phi}_6$	−0.4990	0.0585	−8.5254	0.0000	***
$\hat{\theta}_2$	−0.9051	0.0512	−17.6852	0.0000	***
$\hat{\theta}_6$	0.4790	0.0562	8.5182	0.0000	***
External regressors (mean equation)					
$\hat{\beta}_{XLE}$	0.1050	0.0470	2.2339	0.0255	*
$\hat{\beta}_{USD}$	0.3402	0.1153	2.9492	0.0032	**
Variance equation					
$\hat{\omega}$	0.0265	0.0033	8.0362	0.0000	***
$\hat{\alpha}_1$	−0.0326	0.0217	−1.4974	0.1343	n.s.
$\hat{\beta}_1$	0.9791	0.0009	1125.7638	0.0000	***
$\hat{\gamma}_1$	0.0962	0.0146	6.5962	0.0000	***
Distribution parameters					
skew	0.8083	0.0423	19.0982	0.0000	***
shape (ν)	10.2195	3.0528	3.3476	0.0008	***

Significance: ***$p < 0.001$, **$p < 0.01$, *$p < 0.05$, n.s. – not significant.
GARCH model: eGARCH(1,1). Distribution: sstd.

The four mean parameters ($\phi_2, \phi_6, \theta_2, \theta_6$) are all significant at the 0.1% level under both standard and robust standard errors. The leverage parameter $\gamma_1 = 0.096$ is the single most economically meaningful coefficient: the implied asymmetry is large and statistically very strong ($p < 0.001$ under robust SEs), and it is the largest leverage estimate among the ARMAX group.[2] The ARCH parameter $\alpha_1 = -0.032$ is not significant on its own ($p = 0.1343$): in EGARCH(1,1) this is acceptable because the conditional variance is mostly driven by the persistence term $\beta_1 = 0.979$ and the leverage term γ_1.

The exogenous regressors are jointly significant at the 5% level under robust standard errors, a property that no other model in the ARMAx group achieves: $\hat{\beta}_{XLE}$ ($p = 0.026$) and $\hat{\beta}_{USD}$ ($p = 0.003$). The size of the coefficients also makes economic sense: the once-lagged XLE return enters with a positive sign (energy equities lead the physical market by one day) and the twice-lagged USD return enters with a positive short-run sign of 0.342, which dominates over the very

[1] In M6, the individual Nyblom statistic for α_1 is 0.568, just above the 0.47 critical value at the 5% level. This is a known difficulty in identifying the EGARCH ARCH coefficient separately from the leverage term γ_1; we note this as a caveat without modifying the specification.

[2] In M7, $\gamma_1 = 0.009$ is essentially zero ($p = 0.74$), confirming the empirical fact that the threshold formulation of GJR-GARCH and the log-variance formulation of EGARCH capture different aspects of asymmetry, with the latter being the more flexible parametrisation in the present sample.

short horizon of two trading days.[3]

The Pearson goodness-of-fit test for M6 also shows excellent distributional fit at every grouping (p-values 0.99, 0.999, 0.86 and 0.72 for 20, 30, 40 and 50 groups), the best in the ARMAX family. Putting all of this together: clean diagnostics, joint significance of the exogenous regressors under robust SEs, strong and economically meaningful leverage effect, and stable parameter estimates M6 is the preferred in-sample specification.

6.3 News Impact Curve (NIC) Analysis

The News Impact Curves depicted in Figure 4 provide visual confirmation of the leverage effect documented in Table 12.

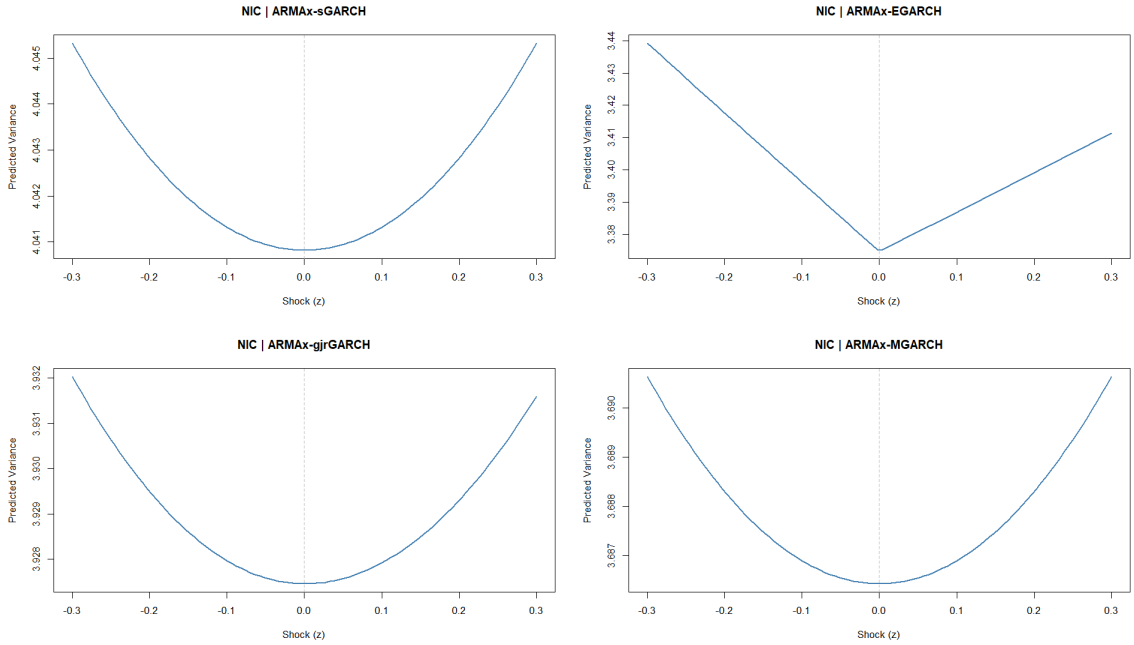


Figure 4: News Impact Curves for ARMAx-GARCH specifications (M5–M8). The curves display the relationship between lagged standardized shocks (x-axis) and predicted conditional variance (y-axis), isolating the asymmetric response of volatility to positive versus negative market movements.

All four ARMAX models exhibit pronounced asymmetry around zero, with a steeper decay in predicted variance following negative shocks compared to positive shocks of equal magnitude. This asymmetry is most pronounced in M6 (ARMAX-EGARCH), where the leverage coefficient $\gamma_1 = 0.097$ captures the market's elevated volatility response to bad news. The EGARCH parametrization proves particularly effective at modeling this asymmetry, as the log-variance formulation allows for flexible curvature. By contrast, the GJR-GARCH specification (M7) exhibits a more muted asymmetric response, consistent with the near-zero leverage estimate $\gamma_1 = 0.009$ in that model. These curves underscore a critical stylized fact in crude oil markets: negative price shocks, driven by geopolitical risks or demand destruction, trigger disproportionately larger

[3]This positive short-run coefficient is not in conflict with the long-run inverse USD/commodity relationship: at the daily frequency, the r_{t-2}^{USD} term picks up momentum-style patterns rather than the long-run co-integration that drives the negative correlation at lower frequencies.

volatility spikes than equivalent positive supply or demand surprises. This inherent asymmetry, well-captured by the EGARCH formulation, reinforces the superiority of the M6 specification for both 80% observations and 80% observations subsets risk forecasting.

6.4 Empirical Results

Figures 5 and 6 display the forecasting performance of the selected M6 specification across three years (2023–2025), decomposed into price-level and log-return dimensions respectively.



Figure 5: WTI price forecasts: ARMAX-EGARCH(1,1) conditional mean and 95% prediction intervals

Price forecasts. The ARMAx mean forecast (blue line) tracks the actual WTI price series (black line) closely throughout all three sub-periods, remaining well within the 95% GARCH confidence band (shaded pink region) at virtually every observation. The band itself is informative: it widens during episodes of elevated realized volatility, and compresses during calmer trending phases.

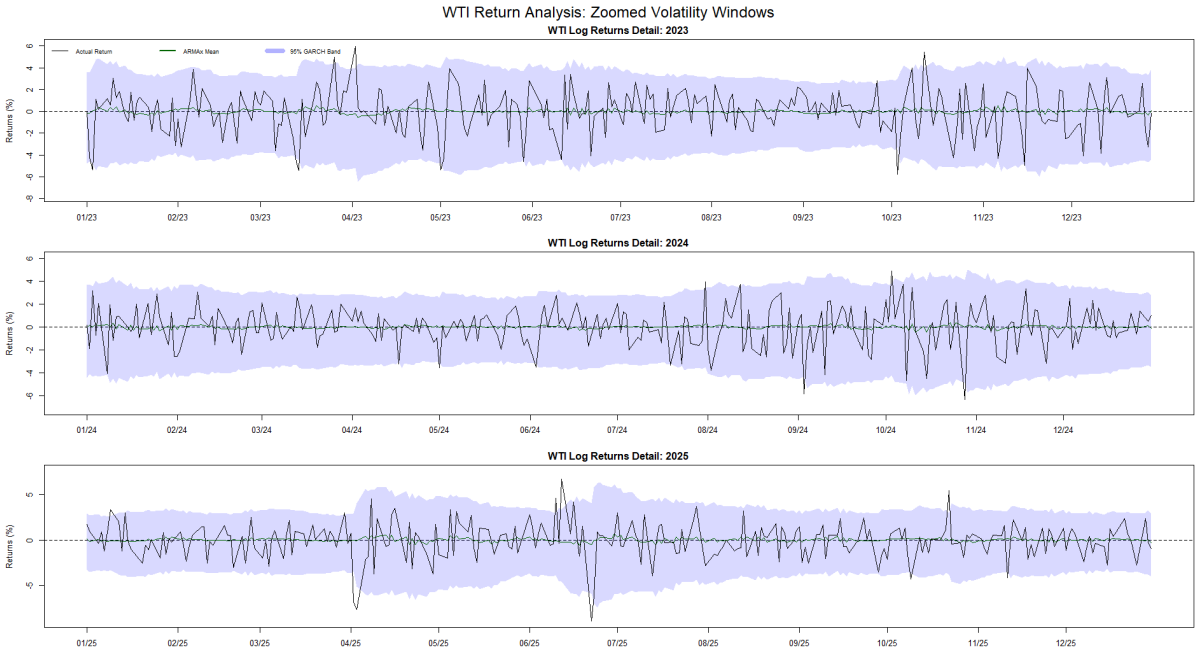


Figure 6: WTI log-return forecasts: ARMAX-EGARCH(1,1) conditional mean and 95% volatility bands

Return forecasts. The log-return panel reinforces this reading. The 95% GARCH band (shaded blue) expands and contracts in response to volatility clustering: the pronounced widening visible in early 2023, around the October 2023 spike, and again in April–July 2025 coincides with the largest realized daily moves in the sample. In all three years the actual return series remains almost entirely within the forecast band, with only a handful of tail exceedances—consistent with the nominal 5% coverage level and with the fat-tailed skewed-t distributional assumption embedded in M6.

7 Conclusion

This paper has addressed two complementary questions on the forecasting of WTI crude oil futures. First, whether a parsimonious ARMAX–GARCH specification properly selected and evaluated can deliver accurate one-step-ahead predictions of WTI returns and volatility; second, whether the inclusion of a set of energy and financial variables can shed light on the leading indicators of crude oil dynamics. A systematic comparison of eight competing ARMAX(6,6)-GARCH(1,1) specifications, all under skewed Student-t innovations, leads to selecting model M6 = ARMAX(6,6)-EGARCH(1,1), which augments the conditional mean with three exogenous regressors the once-lagged Energy Select Sector ETF return (XLE), the once-lagged copper futures return (HG), and the twice-lagged US Dollar Index return (USD). This model is preferred on the basis of:

- clean residual diagnostics, with no remaining ARCH effect at any tested lag and no remaining autocorrelation in the standardised residuals;
- joint significance of the three exogenous regressors at the 5% level under robust standard

errors, a property that no other ARMAx model in our family achieves;

- a strongly significant leverage parameter capturing the well-documented asymmetric volatility response of energy commodities to negative news;
- an excellent Pearson goodness-of-fit across all groupings;
- a competitive forecast accuracy.

The DM test does not statistically discriminate among the eight models on squared mean-forecast errors, so the in-sample considerations have priority. Notably, all four univariate GARCH models (M1–M4) fail the residual variance. The inclusion of exogenous regressors in the mean equation absorbs variance dynamics that the pure GARCH structure could not capture on its own, and is the strongest internal reason to prefer the ARMAX family. The asymmetry of the price interval reflects the fitted left-skewness of the return distribution. On the multivariate side, the joint inspection of the cross-correlation functions, of pairwise Granger causality tests and of a restricted VAR(2) system identifies XLE, the US Dollar Index and copper as the most robust leading indicators of WTI returns at the daily frequency. Refined products futures (RBOB, HEAT) and the S&P 500 co-move contemporaneously with WTI but do not lead it, while carbon emissions, the Baltic Dry Index, gold and (in the most recent portion of the sample) the VIX do not show any robust lead–lag relationship. These findings are fully consistent with standard economic arguments on the role of the dollar, of the energy-equity information channel, and of industrial activity in shaping daily oil returns.

Limitations Several caveats must be acknowledged. First, the analysis is conducted at the daily frequency: some dynamics might be missed at higher (intraday) or lower (weekly, monthly) frequencies. Second, the GARCH family we compare, while broader than the typical "GARCH(1,1) versus GARCH(2,1)" exercise, is still finite, leaving out long-memory variants (FIGARCH, ARFIMA–GARCH) and component -GARCH models. Third, the ARMAX mean equation is linear; nonlinear relationships between WTI and its leading indicators are not modelled. Fourth, potential structural breaks (such as the 2022 Russia–Ukraine energy shock or the OPEC+ supply decisions) are not modelled explicitly. The Nyblom test does not flag any joint instability at the 5% level, but the individual statistic for the EGARCH ARCH term α_1 in M6 is borderline, suggesting some residual identification difficulty in that parameter. Fifth, the 20% test sample contains only 200 observations, which limits the statistical power of the DM test; under longer test samples we would expect the in-sample preference for M6 to translate into a statistically significant OOS edge as well. A further limitation concerns the static nature of the lead-lag structure identified through Cross-Correlation Function analysis. The CCF was computed over the entire sample period, implicitly assuming that the relationships between WTI returns and the selected exogenous indicators (XLE and USD) remain stable across time. However, financial market interdependencies are inherently dynamic and subject to structural change, particularly over a sample period as turbulent as 2022–2026. Had the analysis been conducted over rolling sub-windows, the estimated correlations and optimal lag structures would likely differ across regimes, suggesting that the identified lead-lag relationships may not be stationary.

Directions for further research Natural extensions of this work include:

- enlarging the variance-model family with asymmetric long-memory variants (FIEGARCH, FI-GJR-GARCH) and a component-GARCH structure that disentangles short- and long-run volatility components;
- replacing the linear ARMAx with a multivariate GARCH (DCC-GARCH or BEKK-GARCH) on the multivariable-variable system to model time-varying correlations explicitly;
- augmenting the mean equation with regime-switching or change-point components to accommodate structural shifts in the WTI dynamics;
- complementing point forecasts with full density forecasts and evaluating them through proper scoring rules (CRPS, log-score);
- comparing the ARMAX–EGARCH benchmark established here against machine-learning competitors on the same rolling exercise, in the spirit of Billé et al. (2023);
- Future research could address this limitation by employing time-varying correlation frameworks, such as the Dynamic Conditional Correlation (DCC) model or rolling CCF estimation, to capture the evolving nature of cross-market dependencies.

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